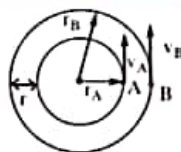
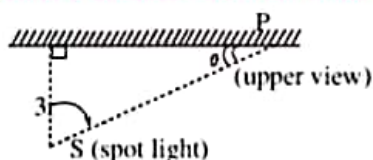


EXERCISE-I*(Kinematics of Circular motion)***LEVEL-I (MAIN)****Straight Objective Type Questions**

- The speed of a motor increases from 1200 rpm to 1800 rpm in 20s. How many revolutions does it make in this period of time ?
1) 400 2) 200 3) 500 4) 800
- Starting from rest a wheel rotates with uniform angular acceleration $2\pi \text{ rad s}^{-2}$. After 4s, if the angular acceleration ceases to act, its angular displacement in the next 4s is
1) $8\pi \text{ rad}$ 2) $16\pi \text{ rad}$ 3) $24\pi \text{ rad}$ 4) $32\pi \text{ rad}$
- A particle 'P' is moving in a circle of radius 'r' with uniform speed v. AB is the diameter of circle and 'C' is the centre. The angular velocity of P about A and C are in the ratio.
1) 1 : 1 2) 1 : 2 3) 2 : 1 4) 4 : 1
- A particle is going parallel to x-axis with constant speed V at a distance 'a' from the axis. Find its angular velocity about an axis passing the origin O, at the instant when radial vector of the particle makes angle θ with the x-axis
1) $\frac{V}{a} \sin^2 \theta$ 2) $\frac{V}{a \sin^2 \theta}$ 3) $\frac{2V}{a} \sin^2 \theta$ 4) $\frac{V \sin^2 \theta}{2a}$
- Two particles A and B are moving on two different concentric circles with different velocities v_A and v_B , then the angular velocity of B relative to A as observed by A is given by:
1) $\frac{v_B - v_A}{r_B - r_A}$ 3) $\frac{v_B - v_A}{r_B + r_A}$
2) $\frac{v_A - v_B}{r_A - r_B}$ 4) $\frac{v_B + v_A}{r_B + r_A}$
- If two particles are moving on same circle or different circles in same plane with different angular velocities ω_1 and ω_2 and different time period T_1 and T_2 in same sense, then the time taken by 2 to complete one revolution w.r.t particle 1 is
1) $T = \frac{T_1 T_2}{T_2 - T_1}$ 2) $T = \frac{T_1 + T_2}{2}$ 3) $T = \frac{T_1 T_2}{T_1 + T_2}$ 4) $T_2 - T_1$

**LEVEL-II (ADVANCED)****Straight Objective Type Questions**

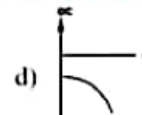
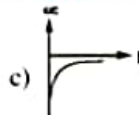
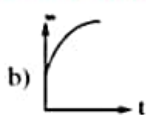
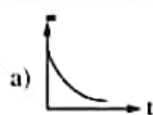
- A spot light S rotates in a horizontal plane with a constant angular velocity of 0.1 rad/s. The spot of light P moves along the wall at a distance 3m. What is the velocity of the spot P when $\theta = 45^\circ$?
a) 0.6 m/s
b) 0.5 m/s
c) 0.4 m/s
d) 0.3 m/s



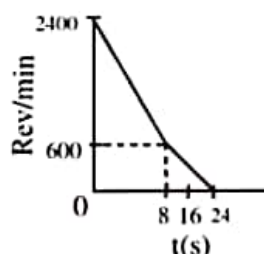
2. A rigid body rotates about a fixed axis with variable angular velocity equal to $\alpha - \beta t$, at time t where α, β are constants. The angle through which it rotates before it stops:

a) $\frac{\alpha^2}{2\beta}$ b) $\frac{\alpha^2 - \beta^2}{2\alpha}$ c) $\frac{\alpha^2 - \beta^2}{2\beta}$ d) $\frac{(\alpha - \beta)\alpha}{2}$

3. A point situated on a wheel decelerates obeying the relation $\omega = \omega_0 - a\theta$ is angular displacement counted from $t = 0$. Which of the following graphs represents the angular acceleration of the wheel.



4. A table fan rotating at a speed of 2400 rpm is switched off and the resulting variation of the rpm with time is shown in the figure. The total number of revolutions of the fan before it come to rest is



- a) 420 b) 280 c) 190 d) 16800

EXERCISE-II

(Dynamics of circular motion)

LEVEL-I (MAIN)

Straight Objective Type Questions

1. A car is moving with a speed of 30 ms^{-1} on a circular path of radius 500 m. If its speed is increasing at the rate of 2 ms^{-2} , the net acceleration of the car is
- 1) 3.6 ms^{-2} 2) 2.7 ms^{-2} 3) 1.8 ms^{-2} 4) 2 ms^{-2}
2. A particle moves in a circle in such a way that, its tangential deceleration is numerically equal to its radial acceleration. If the initial velocity of the particle is V_0 find the variation of its velocity with time

1) $\frac{V_0}{1 + \frac{V_0 t}{R}}$ 2) $\frac{V_0 t}{1 + \frac{V_0 t}{R}}$ 3) $\frac{V_0}{t + \frac{V_0 t}{R}}$ 4) None of above

3. A particle moves in a circular path such that its speed v varies with distance as $V = \alpha\sqrt{s}$ where α is a positive constant. Find the acceleration of the particle after traversing a distance s .

1) $\alpha^2 \sqrt{\frac{1}{4} + \frac{s^2}{R^2}}$ 2) $\alpha^2 \sqrt{\frac{R^2}{4s^2} + \frac{s^2}{R^2}}$ 3) $\alpha \sqrt{\frac{1}{4} + \frac{s^2}{R^2}}$ 4) $\alpha \sqrt{\frac{R^2}{4s^2} + \frac{s^2}{R^2}}$

LEVEL-II (ADVANCED)

More than One correct answer Type Questions

1. A particle of mass m describe path of radius ' r ' and is radial or normal or centripetal acceleration depends on time ' t ' as $a_r = kt^2$. k is +ve constant. Then :
- a) at any time ' t ' force acting on particle is $m\sqrt{kr + k^2 t^4}$
 b) power developed at any time t is $mkrt$
 c) Power developed at any time t is $mk^{3/2}/t^{3/2}$ d) Tangential acceleration is also varying.

Linked Comprehension Type Questions**Passage-I :**

A particle undergoes uniform circular motion. The velocity and angular velocity of the particle at an instant of time is $\vec{v} = 3\hat{i} + 4\hat{j}$ m/s and $\vec{\omega} = x\hat{i} + 6\hat{j}$ rad/sec

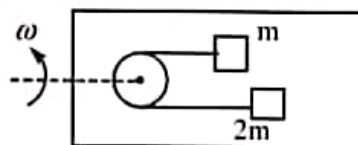
2. The value of x in rad/s is
 a) 8 b) -8 c) 6 d) can't be calculated
3. The radius of circle in meters
 a) 1/2m b) 1m c) 2m d) can't be calculated
4. The acceleration of particle at the given instant is
 a) $-50\hat{k}$ b) $-42\hat{k}$ c) $2\hat{i} + 3\hat{j}$ d) can't be calculated

Passage-II :

A table with smooth horizontal surface is placed in a cabin which moves in a circle of a large radius R . A smooth pulley of small radius is fastended to the table. Two masses m and $2m$ placed on the table are connected through a string over the pulley. Initially the masses are held by a person with the string along the outward radius and then the system is released from the (with respect to the cabin).

5. Find the magnitude of the initial acceleration of the masses as seen from the cabin

- a) $\frac{\omega^2 R}{3}$ b) $\frac{\omega^2 R}{2}$
 c) $\frac{\omega^2 R}{4}$ d) $\omega^2 R$



6. Find the tension in the string.

- a) $\frac{4}{3}m\omega^2 R$ b) $\frac{2}{3}m\omega^2 R$ c) $\frac{1}{3}m\omega^2 R$ d) $m\omega^2 R$

Matrix Matching Type Questions

7. A particle of mass m moving in a circular path of constant radius r such that its centripetal acceleration a_c is varying with time t as $a_c = k^2 r t^2$, where k is a constant.

Column-I

- A) Centripetal force
 B) Tangential force
 C) Power delivered by centripetal force
 D) Power delivered by tangential force

Column-II

- P) mkr
 Q) $mk^2 r^2 t$
 R) $mk^2 r t^2$
 S) zero

8. A particle moves in a circle of radius 1m at a speed of $v = 2t$, where 'v' is in m/sec and 't' in sec. Then match the Columns

Column-I

- A) Radial acceleration of the particle at $t = 1$ sec
 B) Tangential acceleration of the particle at $t = 1$ sec
 C) Tangential acceleration of the particle at $t = 2$ sec
 D) Total acceleration at $t = 1$ sec

Column-II

- P) $2\sqrt{5} \text{ m/s}^2$
 Q) 2 m/sec^2
 R) 3 m/sec^2
 S) 4 m/sec^2

EXERCISE-III

(Applications of Circular Motion)

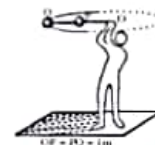
LEVEL-I (MAIN)
Straight Objective Type Questions

1. A boy is sitting on a horizontal platform in the shape of a disc at a distance of 5m from its centre. The boy begins to slip when the speed of wheel exceeds 10 rpm. The coefficient of friction between the boy and platform is ($g = 10 \text{ ms}^{-2}$)

- 1) $\pi^2/6$ 2) $\pi^2/18$ 3) $\pi/6$ 4) $\pi/2$

2. Two spheres of equal masses are attached to a string of length 2m as shown. The string and the spheres are then whirled in a horizontal circle about O at a constant rate. If T_1 is the tension in the string between P & Q and T_2 is the tension in the string between P & O then T_1/T_2 equals. P is connected to the mid point of string.

- 1) 1 2) $\frac{1}{3}$
 3) $\frac{2}{3}$ 4) $\frac{1}{2}$



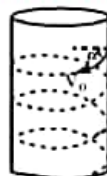
3. A large mass M hangs stationary at the end of a light string that passes through a smooth fixed tube to a small mass m that moves around in horizontal circular path. If l is the length of the string from m to the top end of the tube and θ is angle between this part and vertical part of the string as shown in the fig, then time taken by m to complete one circle is equal to

- 1) $2\pi\sqrt{\frac{l}{g\sin\theta}}$ 2) $2\pi\sqrt{\frac{l}{g\cos\theta}}$
 3) $2\pi\sqrt{\frac{ml}{gM\sin\theta}}$ 4) $2\pi\sqrt{\frac{lm}{gM}}$



4. A particle of mass m moves along the internal smooth surface of a vertical cylinder of radius R as shown. The force which acts on the wall of the cylinder if initially the velocity v_0 of the particle makes an angle α with the horizontal (Assume that particle does not leave contact with the curved surface of the cylinder) is

- 1) $\frac{mv_0^2}{R}$ 2) $\frac{mv_0^2 \cos^2 \alpha}{R}$
 3) $\frac{mv_0^2 \sin^2 \alpha}{R}$ 4) mg



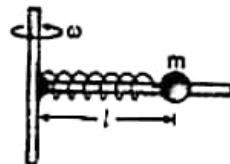
5. An undeformed spring with spring constant k has length l_0 . When the system rotates at an angular velocity ω , the weight with mass m causes an extension in the spring. If l is the length of the rotating spring then

1) $l = \frac{ml_0\omega^2}{k}$

2) $l = l_0 + \frac{ml_0\omega^2}{k}$

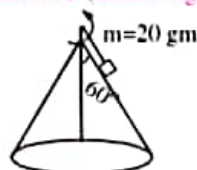
3) $l = \frac{kl_0}{k - m\omega^2}$

4) $l = l_0 - \frac{ml_0\omega^2}{k}$



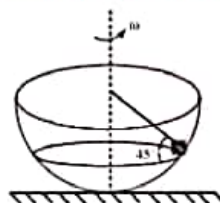
6. A particle is resting on an inverted cone as shown. It is attached to cone by a thread of length 20 cm. String remains parallel to slope of cone. The cone is given angular acceleration of 0.5 rad/sec^2 then at what time does mass leave contact with surface (assuming sufficient friction) :

- 1) 20 sec
2) 10 sec
3) 40 sec
4) 5 sec



7. A hemi spherical bowl is rotated about the axis shown with constant angular velocity ' ω '. A particle is kept as shown at 'P'. If particle also moves with bowl without slipping, then angular velocity is ($g = 10 \text{ m/sec}^2$ and $R = \sqrt{2} \text{ m}$)

- 1) $\sqrt{2} \text{ rad/sec}$
2) $\sqrt{5} \text{ rad/sec}$
3) 3 rad/sec
4) $\sqrt{10} \text{ rad/sec}$



LEVEL-II (ADVANCED)

Linked Comprehension Type Questions

Passage-I :

A block of mass m is kept on rough horizontal turn table at a distance r from centre of table. Coefficient of friction between turn table and block is μ . Now turn table starts rotating with uniform angular acceleration α .

1. Find the time after which slipping occurs between block and turn table

a) $t = \left(\frac{\mu^2 g^2 - \alpha^2 r^2}{\alpha^4 r^2} \right)^{1/4}$ b) $t = \left(\frac{\mu^2 g^2 - \alpha^2 r^2}{\alpha^4 r^2} \right)^{1/3}$ c) $t = \left(\frac{\mu^2 g^2 - \alpha^2 r^2}{\alpha^4 r^2} \right)^{1/2}$ d) $t = \left(\frac{\mu^2 g^2 - \alpha^2 r^2}{\alpha^4 r^2} \right)$

2. Find angle made by friction force with velocity at the point of slipping.

a) $\theta = \tan^{-1}(\alpha t)$ b) $\theta = \tan^{-1}(\alpha t^2)$ c) $\theta = \tan^{-1}(\alpha t^3)$ d) none of the above

Integer Type Questions

3. A car moves with a constant tangential acceleration $a_t = 0.62 \text{ m/sec}^2$ along a horizontal surface circumscribing a circle of radius $R = 40 \text{ m}$. The coefficient of sliding friction between the wheels of the car and the surface is $\mu = 0.2$. The car can ride without sliding upto a distance $12x$ meters if at the initial moment of time its velocity is equal to zero. The value of x is—

HORIZONTAL CIRCULAR MOTION**OBJECTIVE PHYSICS 1A**

4. A car moves uniformly along a horizontal sine curve $y = a \sin (x/\alpha)$ where a and α are certain constants. The coefficient of friction between the wheels and the road is equal to μ . Velocity of the car riding without sliding is $V \leq \alpha \sqrt{\frac{\mu g x}{a}}$. Then $x =$ ———
5. A block of mass ' m ' is kept on a horizontal ruler. The friction coefficient between the ruler and the block is ' μ '. The ruler is fixed at one end and the block is at a distance (L) from fixed end the ruler is rotated about the fixed end in the horizontal plane. If angular speed of the ruler is uniformly increased from zero at an angular acceleration ' α ' when the angular speed be $\omega = \left(\frac{\mu^2 g^2 - L^2 \alpha^2}{L^2} \right)^{2/n}$ the block will slip. Then $n =$ ———?

KEY SHEET (LECTURE SHEET)**EXERCISE-I****LEVEL - I**

1) 3 2) 4 3) 2 4) 1 5) 1 6) 1

LEVEL - II

1) a 2) a 3) c 4) b

EXERCISE-II**LEVEL - I**

1) 2 2) 1 3) 1

LEVEL - II1) a, b 2) b 3) a 4) a 5) a 6) a
7) A-R; B-P; C-S; D-Q 8) A-S; B-Q; C-Q; D-P**EXERCISE-III****LEVEL - I**

1) 2 2) 3 3) 4 4) 2 5) 3 6) 1 7) 4

LEVEL - II

1) a 2) b 3) 5 4) 1 5) 8

(Vertical Circular Motion)

LEVEL-I (MAIN)

Straight Objective Type Questions

1. A 5 kg body is rotated in a vertical circle with a constant speed of 4 ms^{-1} using a string of length 1 m, when the tension in the string is 31N, then the body will be
 - 1) at the lowest point
 - 2) making an angle 30° with vertical
 - 3) at the highest point
 - 4) at horizontal position
2. A simple pendulum of length ' l ' carries a bob of mass m . If the breaking strength of the string is $mg/2$, the maximum angular amplitude from the vertical can be
 - 1) 0°
 - 2) 30°
 - 3) 60°
 - 4) 90°
3. A ball of mass ' m ' is rotated in a vertical circle with constant speed. The difference in tensions at the bottom and horizontal positions would be
 - 1) 6 mg
 - 2) 5 mg
 - 3) 2 mg
 - 4) mg
4. The minimum speed of a bucket full of water whirled in a vertical circle of radius 10 m at the highest point so that the water may not fall is ($g = 10 \text{ ms}^{-2}$)
 - 1) 1 ms^{-1}
 - 2) 2 ms^{-1}
 - 3) 3 ms^{-1}
 - 4) 10 ms^{-1}

5. A stone of mass 1 kg is tied to one end of a string of length 0.5 m. It is whirled in a vertical circle. If the maximum tension in the string is 58.8 N, the velocity at the top is
 1) 1.82 ms^{-1} 2) 2.2 ms^{-1} 3) 3.26 ms^{-1} 4) 2.87 ms^{-1}
6. A body is revolving in a vertical circle with constant mechanical energy. The speed of the body at the highest point is $\sqrt{2rg}$. The speed of the body at the lowest point is
 1) $\sqrt{7gr}$ 2) $\sqrt{6gr}$ 3) $\sqrt{8gr}$ 4) $\sqrt{9gr}$
7. A body is moving in a vertical circle of radius 'r' by a string. If the ratio of maximum to minimum speed is $\sqrt{3}:1$, the ratio of maximum to minimum tensions in the string is
 1) 3 : 1 2) 5 : 1 3) 7 : 1 4) 9 : 1
8. The simple pendulum bob is given a horizontal velocity $\sqrt{gl}$ at the bottom. The angle with the vertical through which the ball swings before its velocity becomes zero is
 1) 120° 2) 60° 3) 45° 4) 30°

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. A mass of 0.1 kg is rotated in a vertical circle using a string of length 20 cm. When the string makes an angle 30° with the vertical, the speed of the mass is 1.5 ms^{-1} . The tangential acceleration of the mass at that instant is
 a) 4.9 ms^{-2} b) 9.8 ms^{-2} c) 2.45 ms^{-2} d) 19.6 ms^{-2}
2. The velocity of a body revolving in a vertical circle of radius 'r' at the lowest point $\sqrt{7gr}$. The ratio of maximum to minimum tensions in the string is
 a) 8 : 1 b) 4 : 1 c) $\sqrt{7}:1$ d) $1:\sqrt{7}$
3. A simple pendulum consisting of a mass M is attached to a string of length "L". Its released from an angle " α ". A pin is located at a distance $T < L$ below the suspension. If ' θ ' is the maximum angle which the string makes after hitting the pin, find ' $\cos \theta$ '
 a) $\frac{L \cos \alpha + l}{L + l}$ b) $\frac{L \cos \alpha + l}{L - l}$ c) $\frac{L \cos \alpha - l}{L - l}$ d) $\frac{L \cos \alpha - l}{L + l}$

More than One correct answer Type Questions

4. A small sphere of mass 'm' suspended by a thread is first taken aside, so that the thread forms the right angle with the vertical and then released. Then which options are correct (Here ' θ ' $\Rightarrow$ is the angle of deflection)
 a) The total acceleration of the sphere as a function of ' θ ' is " $g \sqrt{1 + 3 \cos^2 \theta}$ "
 b) The total tension in the thread as a function of ' θ ' is ' $3mg \cos \theta$ '
 c) The thread tension at the moment when the vertical component of the sphere's velocity is maximum will be $mg\sqrt{3}$
 d) The thread tension at the moment when the vertical component of the sphere's velocity is maximum will be ' mg '

OBJECTIVE PHYSICS IA**WORK-POWER-ENERGY**

5. A small spherical ball is suspended through a string of length ' l '. The whole arrangement is placed in a vehicle, which is moving with velocity ' v '. Now suddenly the vehicle stops and ball starts moving along a circular path. If tension in the string at the highest point is twice the weight of the ball, then
- a) $v = \sqrt{5gl}$ b) $v = \sqrt{7gl}$
- c) Velocity of the ball at highest point is $\sqrt{gl}$ d) Velocity of the ball at highest point is $\sqrt{3gl}$

Matrix Matching Type Questions

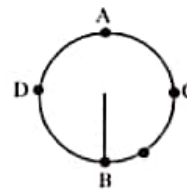
6. One end of string of length 1m is tied to a body of mass 0.5 kg. It is whirled in a vertical circle as shown in figure. If the angular frequency of the body is 4rad/sec.

Column-I

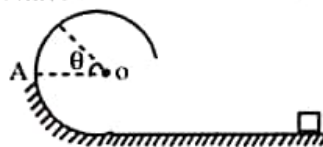
- A) Tension at the lower most point B
B) Tension at the top most point A
C) Tension at point C
D) Tension at point D

Column-II

- P) 12.5 N
Q) 13 N
R) 8 N
S) 3 N

**Integer Type Questions**

7. A circular track of radius 1m as shown in the given figure. A small block is moving on horizontal part of the track with speed $v_0 = \sqrt{44} \text{ m/s}$. The value of $\sin \theta$ at which it leaves the track is $x/5$. Find the value of ' x '.



8. A smooth hollow fixed hemisphere has a radius R of 23m. A vertical smooth circular track of centre O is as such $OC = r$. The block is released from rest from point A . The minimum value of ' r ', so that block complete its turn about O is $3n$ m. Find the value of n (take $\theta = 37^\circ$)

